

Homework - 6

1. Let K be the splitting field of a polynomial $g(x) \in k[x]$ and $f(x) \in k[x]$ be irreducible. Suppose $f(x) = p(x) \cdot q(x) \cdots$, where p & q are irreducible in $K[x]$. Prove that $\exists$ a k -automorphism $\sigma: K \rightarrow K$ such that $\sigma(p(x)) = q(x)$.
2. Let E/k be an algebraic extⁿ, $\alpha, \beta \in E$. Prove that $\exists \sigma: E \rightarrow E$, a k -automorphism with $\sigma(\alpha) = \beta \iff \text{irr}(\alpha; k) = \text{irr}(\beta; k)$.
3. Let K be the splitting field of a polynomial $f(x) \in k[x]$ and let $g(x) \in k[x]$ be irreducible. Assume that $g(x)$ has a root in K . Prove that $g(x)$ splits over K .
4. Let m be an integer that is not a square and let $\alpha = a + b\sqrt{m}$, $a, b \in \mathbb{Q}$, be a root of a polynomial $f(x) \in \mathbb{Q}[x]$. Show that $a - b\sqrt{m}$ is also a root of $f(x)$.
5. Give an infinite field K of positive characteristic p such that the map $\varphi: K \rightarrow K$, $x \mapsto x^p$ is not surjective.